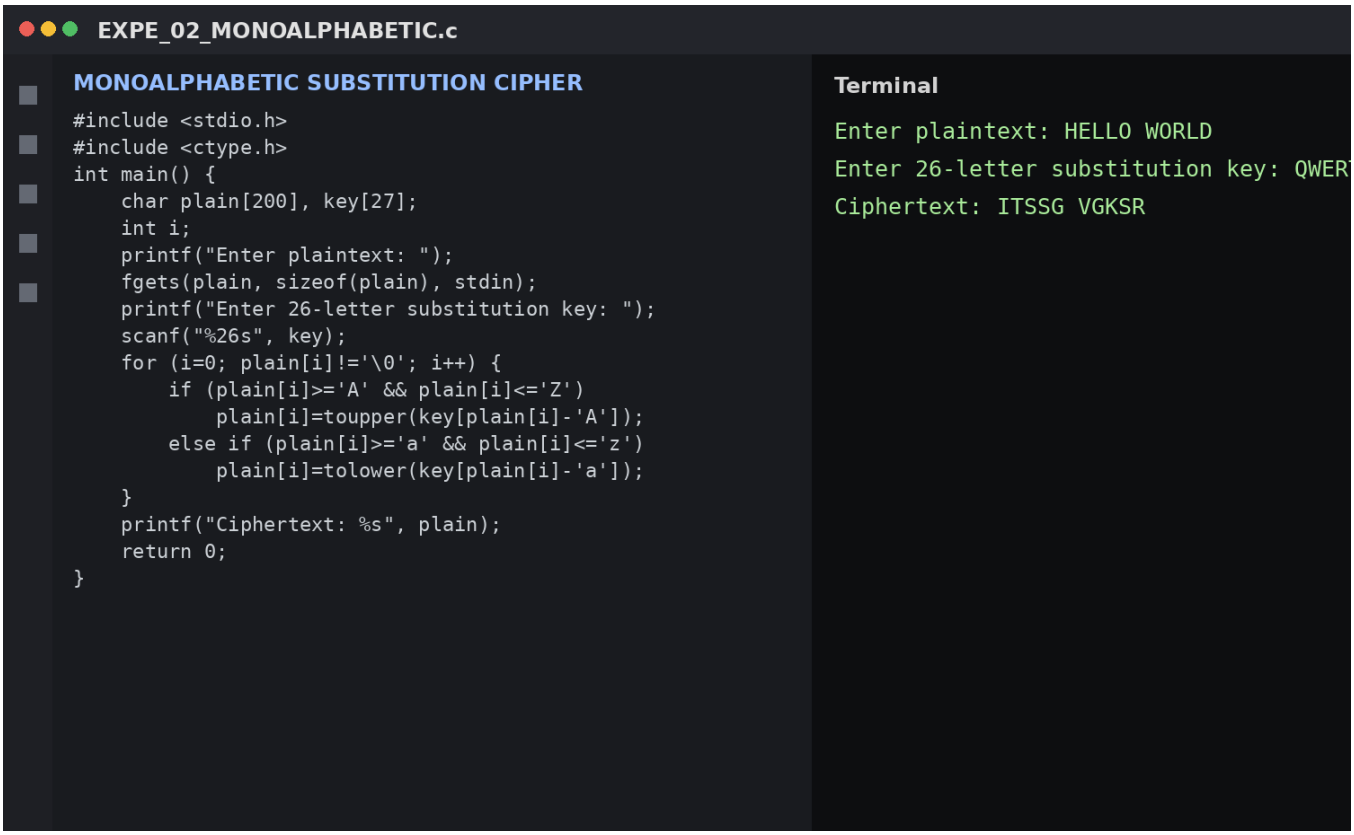
EXPERIMENT – MONOALPHABETIC SUBSTITUTION CIPHER

C Program

```
#include <stdio.h>
#include <ctype.h>
int main() {
    char plain[200], key[27];
    int i;
    printf("Enter plaintext: ");
    fgets(plain, sizeof(plain), stdin);
    printf("Enter 26-letter substitution key: ");
    scanf("%26s", key);
    for (i=0; plain[i]!='\0'; i++) {
        if (plain[i]>='A' && plain[i]<='Z')
            plain[i]=toupper(key[plain[i]-'A']);
        else if (plain[i]>='a' && plain[i]<='z')
            plain[i]=tolower(key[plain[i]-'a']);
    }
    printf("Ciphertext: %s", plain);
    return 0;
}
```

Execution Screenshot



Sample Output

```
Enter plaintext: HELLO WORLD
Enter 26-letter substitution key: QWERTYUIOPASDFGHJKLZXCVBNM
Ciphertext: ITSSG VGKSR
```